

The Two-Tower Recommender

Matching the right **article** to the right **agency**

A visual, plain-language tour of how two neural "towers" learn to predict whether a story belongs with an agency
— scored from 0 to 1.

Article features

Agency profile

Collaborative filtering

Match score

● THE CHALLENGE

Too many stories, too many agencies

Hand-matching content to the right outlet is slow, inconsistent, and impossible at scale.

Thousands of articles

New stories arrive every hour across every topic and region.

Hundreds of agencies

Each has its own focus, audience, region and appetite.

Manual & slow

Editors can't review every possible pairing by hand.

The goal

Auto-score every (article, agency) pair from 0 to 1.

We want a model that, for any story and any agency, instantly answers: “How likely is this a good match?”

Think of it as a **matchmaker**

1. Two profiles

The article describes itself; the agency describes itself.

2. Two fingerprints

The model turns each profile into a compact list of numbers.

3. Compare

It checks how well the two fingerprints fit together.

4. A match score

Out comes a number from 0 (no) to 1 (perfect fit).

Just like a dating app compares two people, the two-tower model compares a story and an agency — and predicts the chance they belong together.

- FIRST, A KEY CONCEPT

What is an **embedding**?

A way to turn words and features into coordinates on a map, where similar things sit close together.

Meaning becomes location

"Human rights in Brazil" lands near "Politics in LATAM" — far from "Pacific Rim chips".

A list of numbers

Each item becomes a vector, e.g. [0.21, -0.88, 0.04, ...] — its fingerprint.

Closeness = relevance

If two fingerprints are near each other, the items are related.

An embedding tower is simply the part of the model that produces these fingerprints.

- THE ARCHITECTURE

Two towers, built **independently**

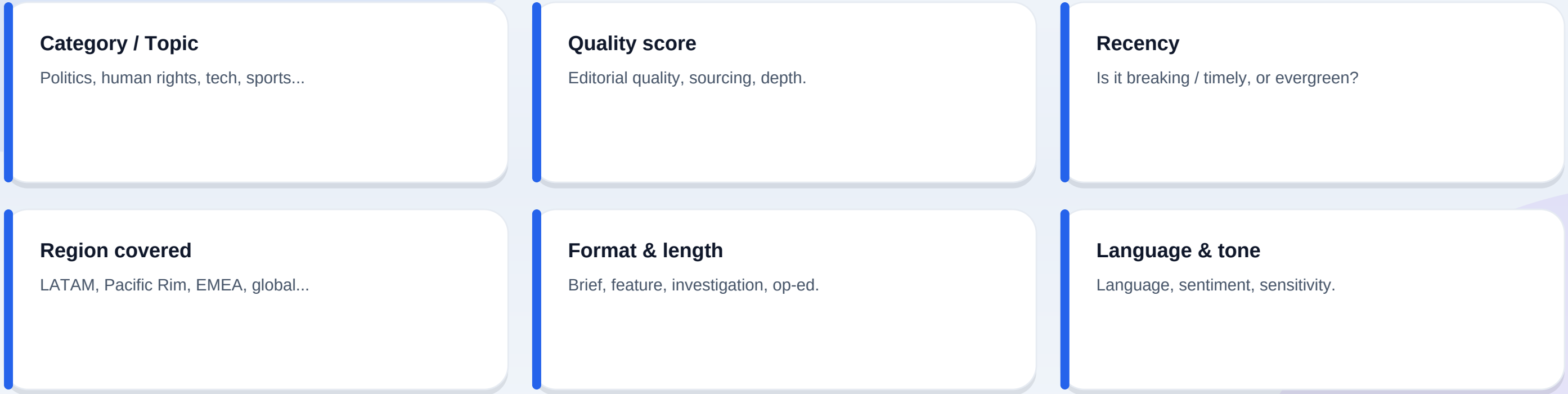
Each tower is its own small neural network. They never share inputs — they only meet at the very top.



Because the towers are separate, we can pre-compute agency fingerprints once and reuse them — making real-time scoring extremely fast.

The Article Tower

It takes everything we know about the story and compresses it into one blue fingerprint.



These features flow up the tower — out comes the article embedding.

The Agency Tower

An agency is best described in words — so this tower reads a short profile and structured facts.

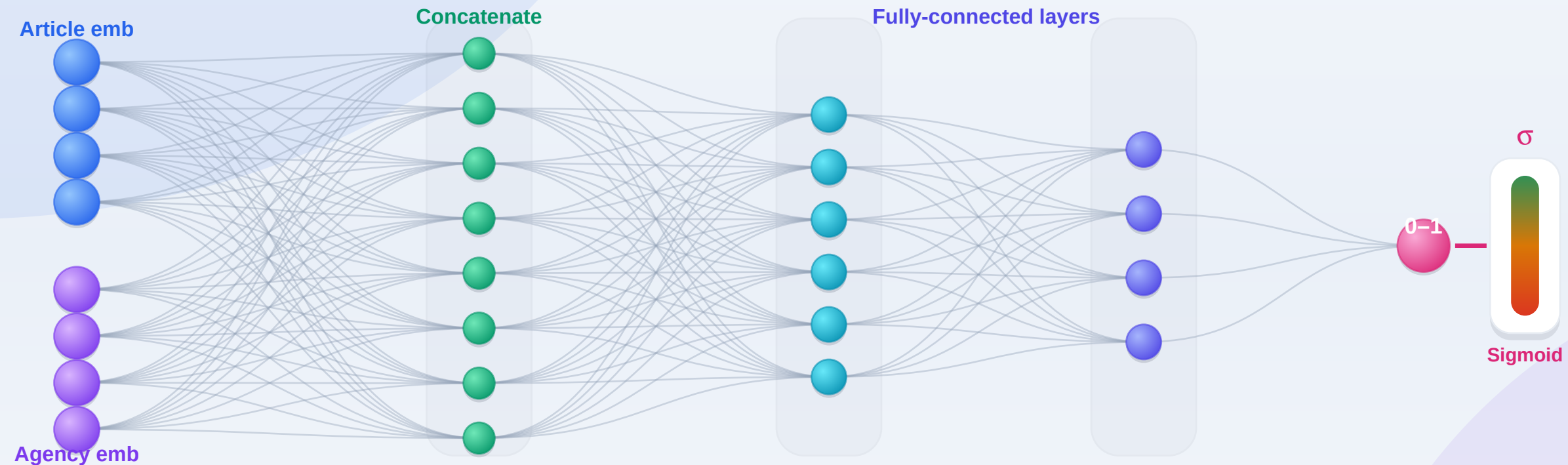
“Independent agency focused on politics and human rights, with strong coverage across LATAM. High weekly publishing capacity and a long, reliable track record with us.”

Topical interests Politics, human rights, climate...	Regions covered LATAM, Pacific Rim, EMEA...	Capacity How much they can publish.	History with us Relationship length & reliability.
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A text encoder turns the paragraph into numbers, combined with the structured facts — the agency embedding.

Where the towers **meet**

Join the two fingerprints, pass them through a fully-connected network, and squeeze the result into a 0–1 score.



Concatenate → fully-connected (dense) layers → sigmoid → a probability between 0 and 1.

Classic two-tower setups compare fingerprints with a dot-product for ultra-fast retrieval; this dense interaction head trades a little speed for sharper ranking accuracy.

Adding collaborative filtering

Content features describe what things are. Collaborative filtering learns from who interacted with what.

ID embeddings (hybrid)

Give every agency & article a learned ID vector, concatenated with the content fingerprint — capturing behavior the words don't.

History pooling

Average the embeddings of articles an agency previously accepted to build a "taste vector".

Graph / GNN signal

Model the agency-article interaction graph (e.g. LightGCN) and learn embeddings from the connections themselves.

Solves cold-start

Content handles brand-new items; collaborative signal sharpens matches for familiar ones. Best of both.

The winning recipe is usually hybrid: content + collaborative in the same two towers.

• HOW IT LEARNS

Teaching the model from **past matches**

Positive & negative pairs

Pairs that were accepted -> label 1. Pairs ignored or never shown (sampled) -> label 0.

Binary cross-entropy

The model is penalised whenever its predicted score is far from the true 0/1 label.

Trained jointly

Both towers and the scoring head learn together via gradient descent, sharpening every fingerprint.

Over millions of historical article-agency interactions, the towers learn to place good matches close together and bad matches far apart.

• A WORKED EXAMPLE

One story, scored against three agencies

INCOMING ARTICLE

“Crackdown on press freedom across LATAM”

Politics

Human rights

LATAM

Breaking

High quality

Agency A - LATAM Politics & Rights desk

Politics · human rights · LATAM · high capacity

0.93

Recommend

Agency B - Global human-rights review

Human rights · global · medium capacity

0.71

Consider

Agency C - Pacific Rim tech & business

Technology · business · Pacific Rim

0.08

Skip

The model ranks agencies instantly — editors act on a short, confident shortlist.

• WHY IT MATTERS

The business **payoff**

Scale & speed

Millions of pairings scored in milliseconds — far beyond manual reach.

Better relevance

Sharper matches lift acceptance and audience engagement.

Cold-start ready

Content features score brand-new stories and agencies from day one.

Frees editors

People focus on judgment calls, not screening every pair.

Self-improving

Every new interaction makes the next recommendation better.

Consistent & auditable

Data-driven scores replace ad-hoc, subjective routing.

• IN A NUTSHELL

Key takeaways

1

Two independent towers turn an article and an agency into comparable numeric fingerprints (embeddings).

2

A fully-connected head with a sigmoid joins the fingerprints and outputs a 0-1 match score.

3

Mixing content features with collaborative filtering captures both meaning and behavior - and beats cold-start.

4

Pre-computed agency fingerprints make it fast, scalable, and self-improving in production.

The right story reaches the right agency — automatically, and at scale.

Presentation by Omar Kamal Hosney · omkamal.github.io